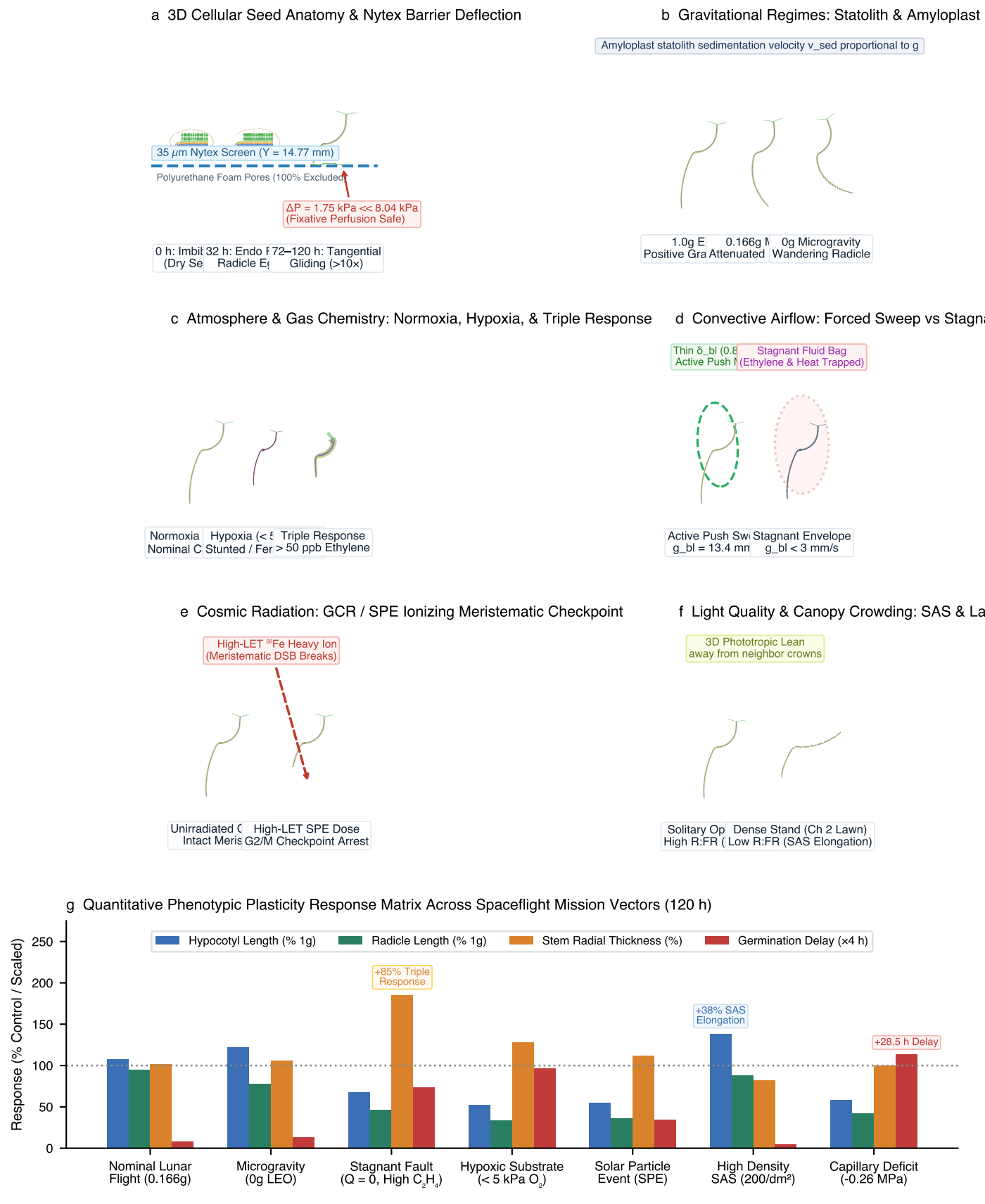

a 3D Cellular Seed Anatomy & Nytex Barrier Deflection
35 μm Nytex Screen (Y = 14.77 mm)
Polyurethane Foam Pores (100% Excluded)
ΔP = 1.75 kPa << 8.04 kPa
(Fixative Perfusion Safe)
0 h: Imbil
(Dry Se
32 h: Endo
Radicle E
72–120 h: Tangential
Gliding (>10×)
b Gravitational Regimes: Statolith & Amyloplast Se
Amyloplast statolith sedimentation velocity v_sed proportional to g
1.0g E
Positive Gra
0.166g
Attenuated
0g Microgravity
Wandering Radicle
c Atmosphere & Gas Chemistry: Normoxia, Hypoxia, & Triple Response
Normoxia
Nominal C
Hypoxia (<
Stunted / Fer
Triple Response
> 50 ppb Ethylene
d Convective Airflow: Forced Sweep vs Stagnan
Thin δ_bl (0.8
Active Push
Stagnant Fluid Bag
(Ethylene & Heat Trapped)
Active Push Sw
g_bl = 13.4 mn
Stagnant Envelope
g_bl < 3 mm/s
e Cosmic Radiation: GCR / SPE Ionizing Meristematic Checkpoint
High-LET 56Fe Heavy Ion
(Meristematic DSB Breaks)
Unirradiated C
Intact Meris
High-LET SPE Dose
G2/M Checkpoint Arrest
f Light Quality & Canopy Crowding: SAS & Late
3D Phototropic Lean
away from neighbor crowns
Solitary Op
High R:FR (
Dense Stand (Ch 2 Lawn)
Low R:FR (SAS Elongation)
g Quantitative Phenotypic Plasticity Response Matrix Across Spaceflight Mission Vectors (120 h)
Hypocotyl Length (% 1g)
Radicle Length (% 1g)
Stem Radial Thickness (%)
Germination Delay (×4 h)
Response (% Control / Scaled)
0
50
100
150
200
250
+85% Triple Response
+38% SAS Elongation
+28.5 h Delay
Nominal Lunar Flight (0.166g)
Microgravity (0g LEO)
Stagnant Fault (Q = 0, High C2H4)
Hypoxic Substrate (< 5 kPa O2)
Solar Particle Event (SPE)
High Density SAS (200/dm²)
Capillary Deficit (-0.26 MPa)